LHμC Booster

Max Topp-Mugglestone

max@topp-mugglestone.co.uk

Overview

- Concept
  - Accelerator options
- Fixed Field Accelerators
- Vertical Fixed Field Accelerators
- Application to LHμC
- Conclusions

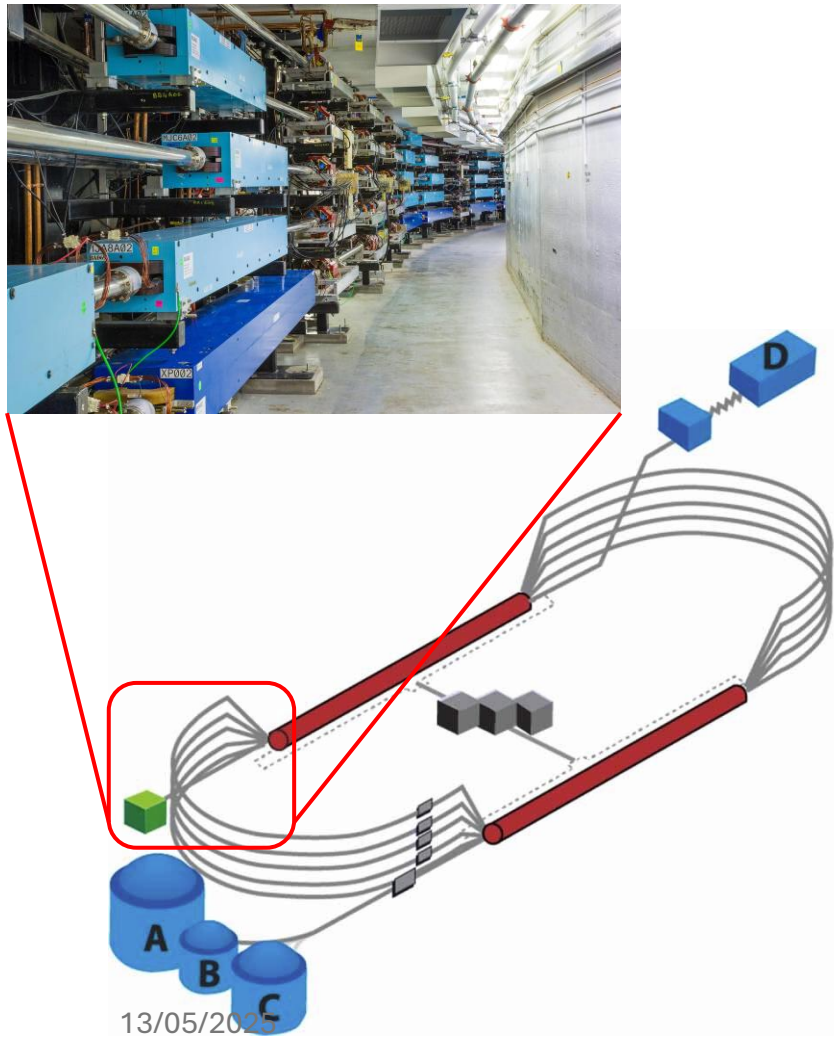
Booster concept

- Racetrack RLA
- Boost energy of muon beams from 50 GeV to 500 GeV or 1 TeV

Considerations:

- Large energy factor (~10-20x gain)
  - Require order 10s of turns for efficient acceleration
- Rest lifetime of muon is 2.2μs
  - Acceleration cycle must be sub-millisecond order (accounting for relativistic effects)

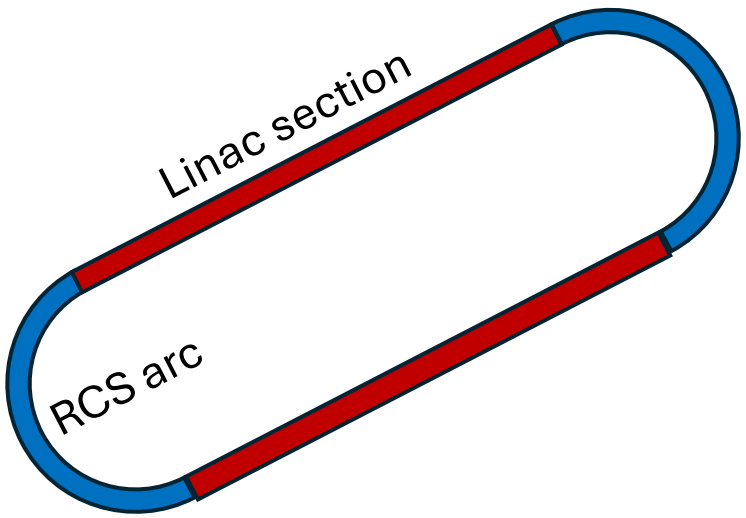
Booster concept



Pictured: CEBAF racetrack RLA
Separated-arc design: 5 passes through individual arcs and shared linac sections (with splitter/recombiner sections) to match between arcs and linacs

For LHμC ?
Large number of turns: large number of individual arcs (expensive) and complicated splitter designs – separated-arc concept may not be optimal for LHμC

Booster concept

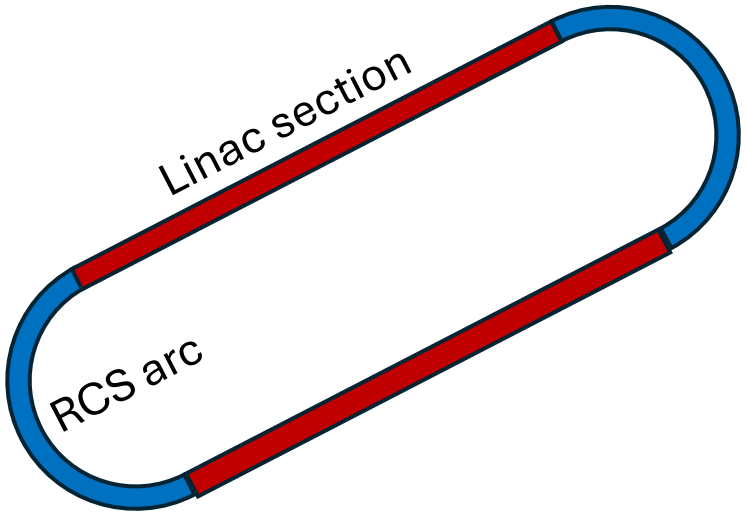


Rapid-cycling synchrotron (RCS) option:
Single (or small number of) arcs
Increase magnetic field proportional to beam energy

Able to accelerate reasonable energy range/number of turns in single arc (less expensive/complex than separated-arc design)
Ramping optics and dipole fields synchronously with beam energy means same path length* and same optics for all energies

*not true for hybrid RCS

Booster concept



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For LHμC ?
Muon decays demand rapid acceleration – ramping magnets within required timeframr is challenging

Booster concept

Is there a solution with:

- **No requirement for individual arcs for each energy**
- **No ramped magnets**
- **Consistent path length**

All in one machine type?

Fixed Field Accelerators

Time-independent magnetic fields means…

- No ramp times
  - Rate of acceleration limited only by RF
  - Mitigates engineering challenges of designing and powering fast-ramping dipoles
- All magnets can be superconducting
- Orbit position moves as a continuous function of energy (single arc, multiple energies)

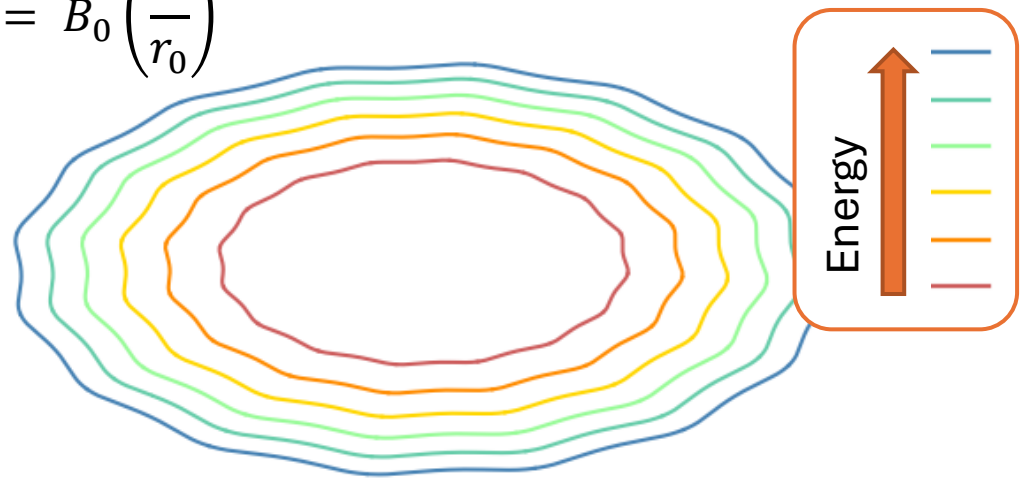
Spatial dependence of magnetic fields allows tune control

"Conventional" horizontal-excursion FFA (hFFA):

Orbits move outwards with increasing energy

Fields increase radially

$$B(r) = B_0 \left(\frac{r}{r_0}\right)^k$$



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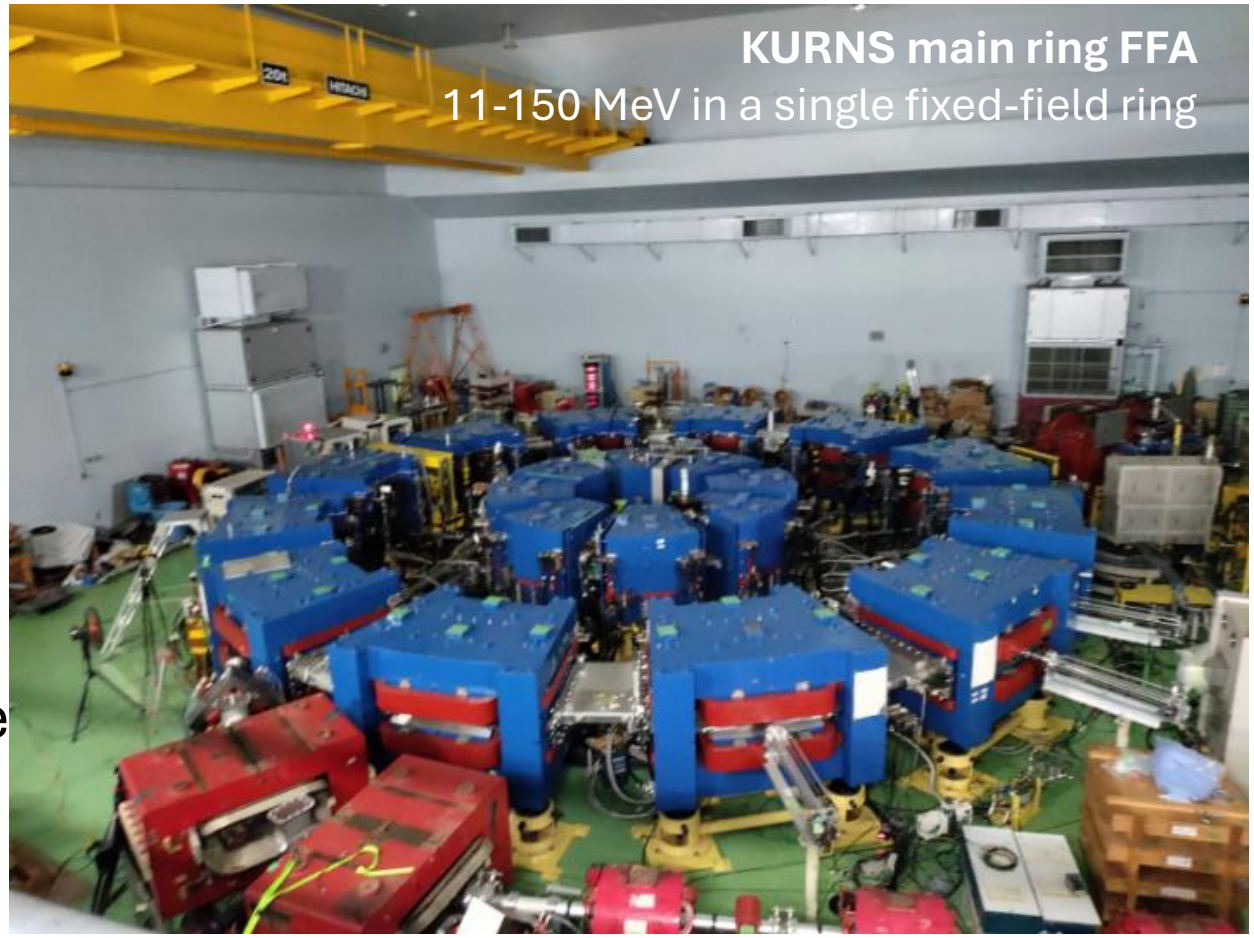


Vertical-Excursion Fixed Field Accelerators

Vertical-excursion FFA (vFFA):

- Higher energy orbits are vertically translated copies of lower energy orbits
- Zero chromaticity if fields increase with vertical coordinate (Z) following scaling law

$$B = B_0 e^{mZ}$$

Zero path length difference means…

→ Zero momentum compaction factor α_c

→ Transitionless

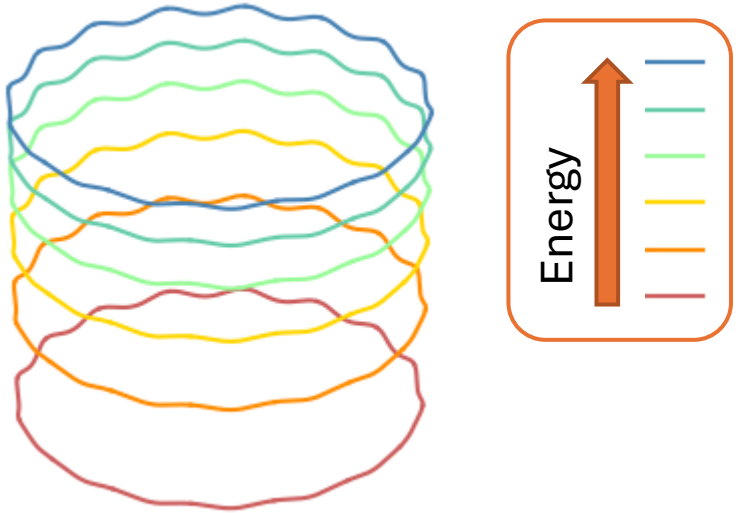
→ Quasi-isochronous for relativistic particles

→ on-crest acceleration can be used

Vertical-excursion FFA (vFFA):

Orbits move upwards with increasing energy

Fields increase vertically



Properties of the vFFA

Non-planar closed orbit

Intrinsically coupled optics

Scaling laws → optics determined by closed orbit and field index *m*

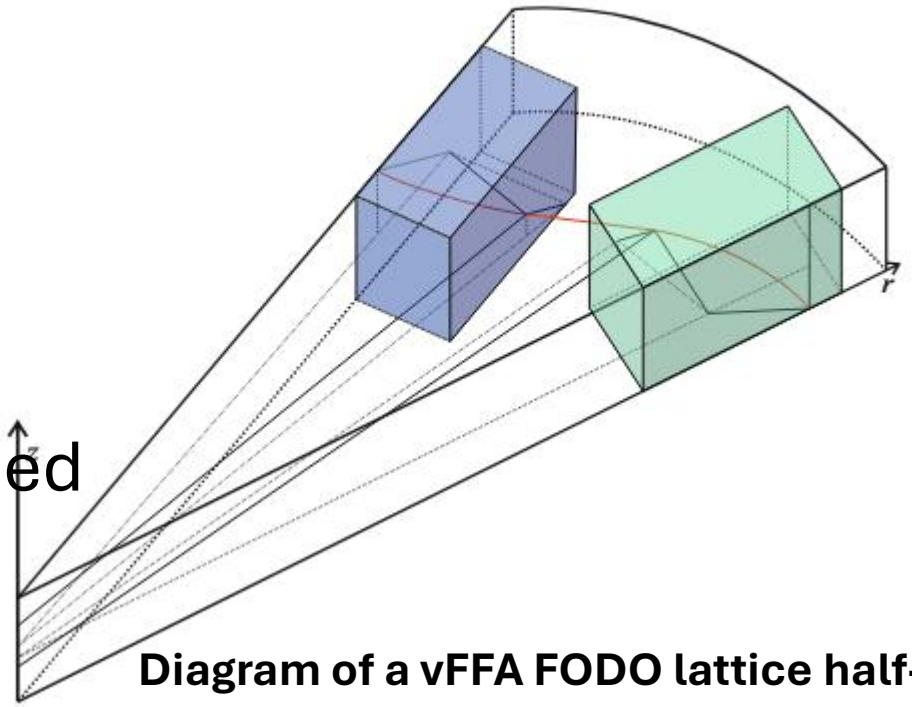


Diagram of a vFFA FODO lattice half-cell

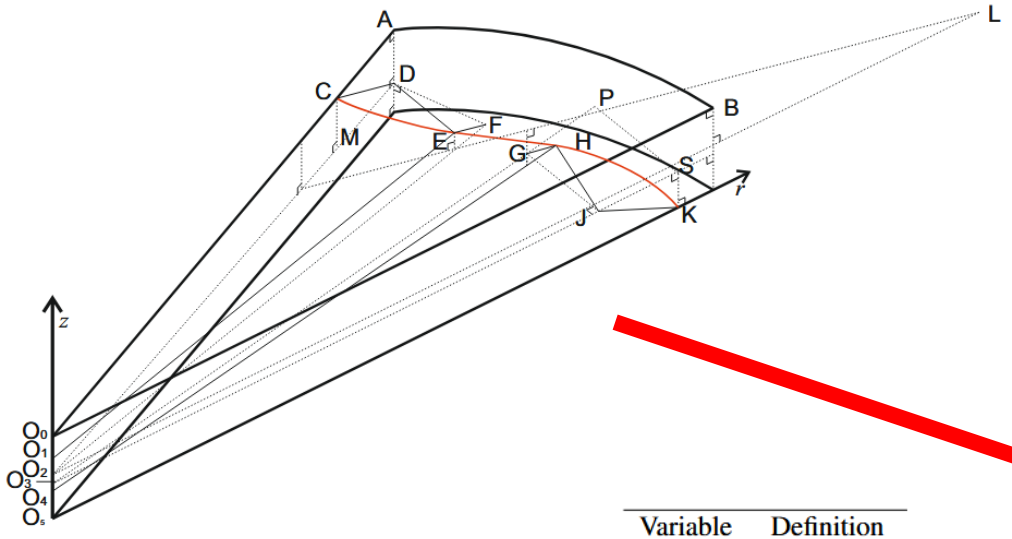
F-magnet is shaded in green; D-magnet is shaded in blue.

Analytic model of vFFA has been developed

"Scaling Fixed Field Accelerators: theory and modelling of horizontal- and vertical-excursion accelerators" – PhD thesis 2024

Properties of the vFFA

Closed orbit



Parameter	Definition
β_F	$\angle SO_3P$
β_D	$\angle DO_2F$
$\frac{\pi}{N}$	$\angle AO_0B$
r_0	$\overline{O_5K}$
θ_F	$\angle HJK$
γ_F	$\angle SJK$

Variable	Definition
θ_D	$\angle CDE$
ρ_F	$\overline{HJ} = \overline{JK}$
ρ_D	$\overline{CD} = \overline{ED}$
r_1	$\overline{O_4H}$
r_2	$\overline{O_1E}$
r_3	$\overline{O_0C}$
L_s	$\overline{EH}$
γ_D	$\angle CDM$

Geometric constraints can be used to determine closed orbit from input parameters

Some important variables for each magnet: ρ, γ, r (radius of curvature, inclination of plane of curvature, radius wrt machine centre)

$$\theta_D = \arctan\left(\frac{\sqrt{\left(\tan\frac{\pi}{N} - \cos\gamma_F \tan\theta_F\right)^2 + \sin^2\gamma_F \sin^2\theta_F \left(\tan\frac{\pi}{N}\cos\gamma_F \tan\theta_F + 1\right)^2}}{\sqrt{1 - \sin^2\gamma_F \sin^2\theta_F}\left(\tan\frac{\pi}{N}\cos\gamma_F \tan\theta_F + 1\right)}\right),$$

$$\sin\gamma_D = \frac{\sin\theta_F}{\sin\theta_D}\sin\gamma_F,$$

$$\rho_F = r_0 \frac{\tan\beta_F}{\sin\theta_F + (1 - \cos\theta_F)\cos\gamma_F \tan\beta_F},$$

$$r_1 = \rho_F \frac{\sin\theta_F}{\sin\beta_F},$$

$$r_2 = r_1 \frac{\cos\beta_F + \tan\theta_F \cos\gamma_F \sin\beta_F}{\cos\left(\frac{\pi}{N} - \beta_D\right) + \tan\theta_F \cos\gamma_F \sin\left(\frac{\pi}{N} - \beta_D\right)},$$

$$\rho_D = r_2 \frac{\sin\beta_D}{\sin\theta_D},$$

$$r_3 = r_2 \cos\beta_D - \rho_D\left(1 - \cos\theta_D \cos\gamma_D\right).$$

Properties of the vFFA

Magnet body Hamiltonian

$$\mathcal{H} \simeq \frac{p_x^2}{2} + \frac{p_z^2}{2} - \frac{1}{\rho + \frac{\sin\gamma}{m}} \left[\cos\gamma \left(m + \frac{2\sin\gamma}{\rho} \right) xz - \frac{1}{2} m \left(x^2 - z^2 \right) \sin\gamma \right] + \frac{1}{\rho\left(\rho + \frac{\sin\gamma}{m}\right)} \left(x^2 \cos^2\gamma + z^2 \sin^2\gamma \right)$$

x, z : horizontal and vertical transverse coordinates

m : normalised field gradient

ρ: radius of curvature

γ : inclination (angle of the plane of curvature in magnet)

m is a constant around the ring

→ Sign of focusing terms can only be changed if sign of ρ changes

→ reverse bends are required

Hamiltonian has normal quad + skew quad + geometric terms

Coupling in the vFFA

Magnet body Hamiltonian

$$\mathcal{H} \simeq \frac{p_x^2}{2} + \frac{p_z^2}{2} - \frac{1}{\rho + \frac{\sin\gamma}{m}} \left[\cos\gamma \left(m + \frac{2\sin\gamma}{\rho} \right) xz - \frac{1}{2} m \left(x^2 - z^2 \right) \sin\gamma \right] + \frac{1}{\rho\left(\rho + \frac{\sin\gamma}{m}\right)} \left(x^2 \cos^2\gamma + z^2 \sin^2\gamma \right)$$

geometric terms – always present
skew quad term – proportional to $\cos\gamma$
normal quad term – proportional to $\sin\gamma$

If $\gamma = 90°$, we can remove the coupling in this Hamiltonian!

However... cells need at least 2 magnets.

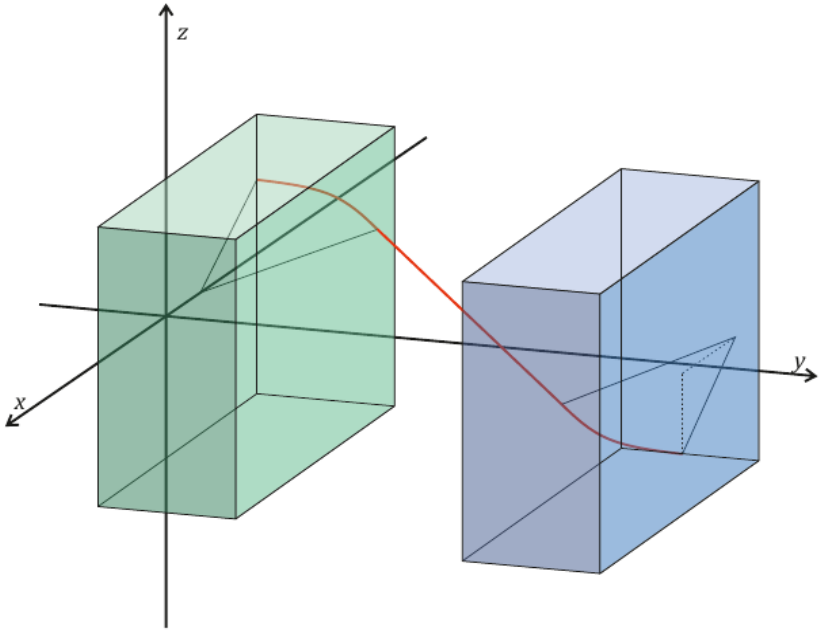
Closed orbit equations

$$\sin\gamma_D = \frac{\sin\theta_F}{\sin\theta_D} \sin\gamma_F,$$

$\boldsymbol{\gamma_F = \gamma_D}$ **only if** $\boldsymbol{\theta_F = \theta_D}$

Decoupled insertions

- Inclination of both magnets can be 90° if the net bending angle of the cell is zero



- A straight insertion with a 90° inclination enables the complete negation of coupling over the length of the straight

Decoupled insertions

When optics are decoupled:

- **Orbit can be kicked/bumped in one plane without affecting the other**

  → Injection and extraction design becomes much less restrictive

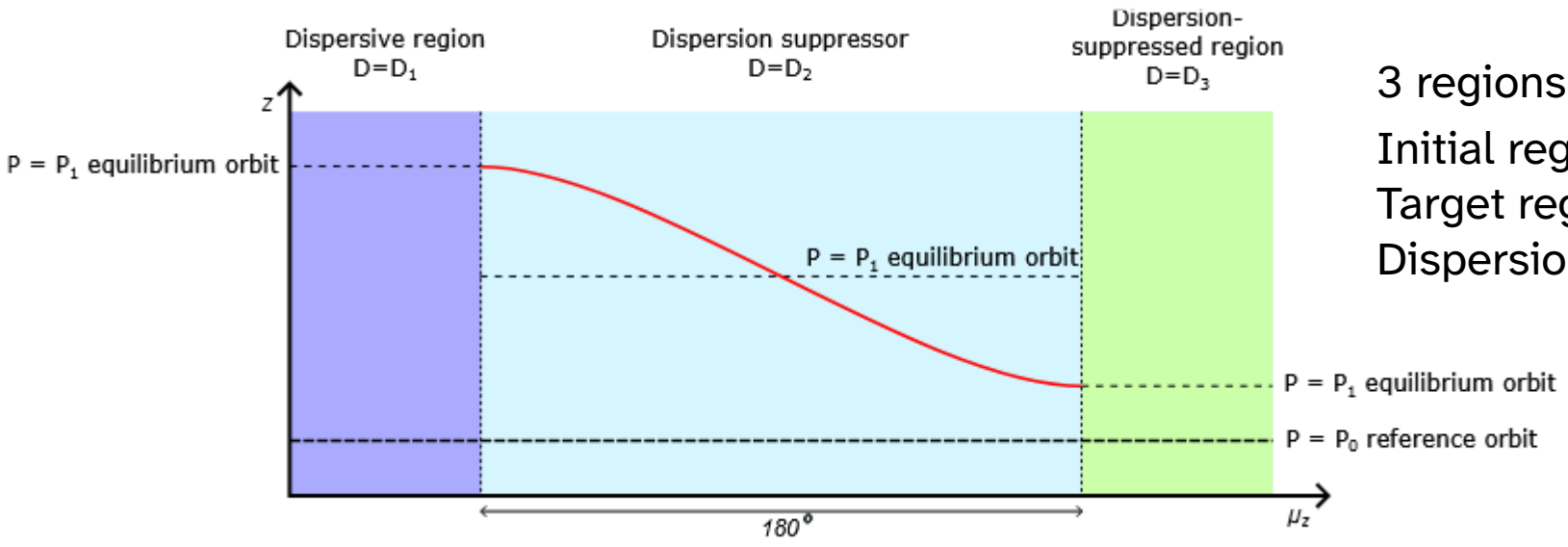
- **Betatron oscillations can be induced in the dispersive plane**

  → Possibility of mitigating transverse/longitudinal issues from RF by tuning cell phase advance between cavities

  → Enables design of dispersion suppressors

Dispersion suppression

In a decoupled straight:



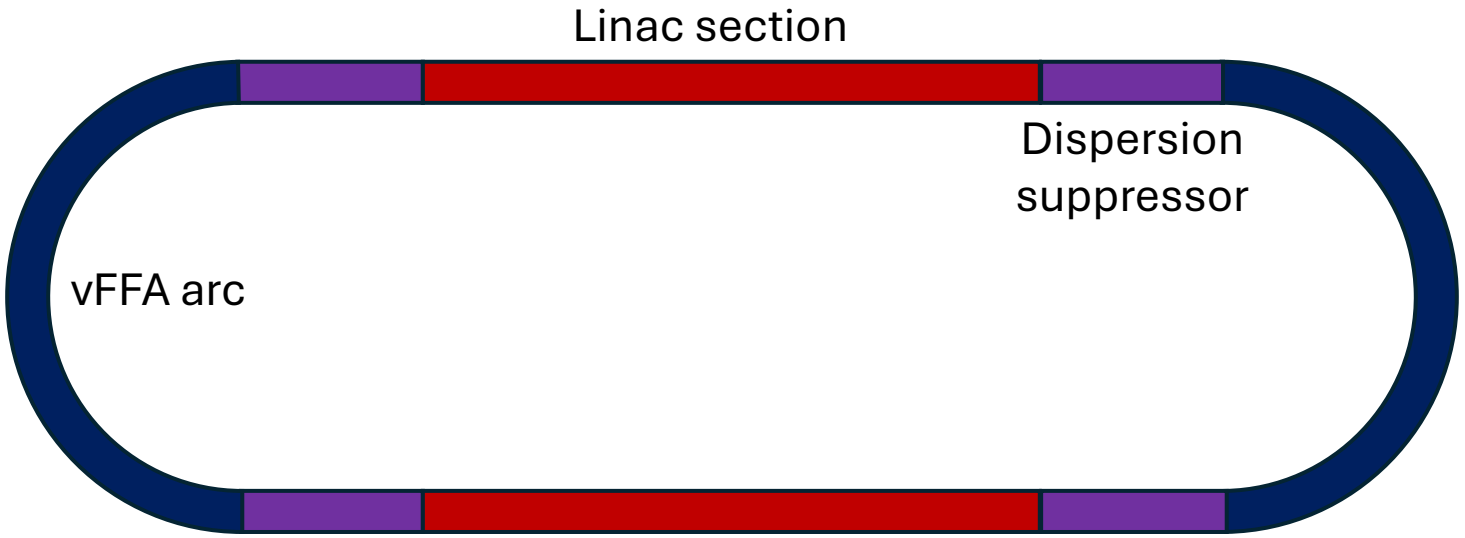
3 regions with different dispersions $D_Z = \frac{1}{m}$

Initial region: D_1 is at nominal value for arcs

Target region : D_3 is at desired value (can be zero)

Dispersion suppressor : $D_2 = (D_1 + D_3)/2$

vFFA for LHμC



- Racetrack RLA with:
  - vFFA arcs
  - Dispersion-suppressed straights

Fixed-field arcs with no need to ramp magnets

- **No constraints** on **acceleration rate**

Fixed path length

- No changes in **time-of-flight** over acceleration

vFFA parameter study

- **LHμC** arc is comparable to RCS1 of **Muon Collider** study
- vFFA alternative to RCS1 has been proposed as part of MC collaboration
  - ~6km circumference
  - 63 GeV to 300 GeV acceleration
  - Peak fields <7 T
  - Drift lengths >1 m

	RCS1	Stage 1 vFFA
Circumference [m]	5990	5990
Injection Energy [TeV]	0.06	0.06
Extraction Energy [TeV]	0.3	0.3
NC Ramped Magnets	Yes	No
SC Fixed magnets	No	Yes
Ramp Rate [T/s]	4200	0
Vertical Excursion [m]	0	0.048
Relative path length difference	0.0	0.0
Peak Dipole Field On Orbit [T]	1.8 (NC)	6.93
Peak Dipole Field (Good Field Region) [T]		12.52
Drift length [m]		1.18
Tune		(0.382, 0.079)

- Parameter study of vFFA offer reasonable alternative to RCS with many potential advantages

Conclusions

- **Racetrack vFFA** is a promising option for the **LHμC booster** RLA
  - No ramped magnets
  - Single arc for multiple energies
  - Constant path length over acceleration
- **MuCol parameter studies** show good performance of **vFFA** in similar parameter space
- **Next steps**:
  - Dedicated parameter study for **LHμC**
  - 6D Simulation of regular vFFA lattices has been started
  - Simulation of vFFA lattices with decoupled insertions and dispersion suppressors must be proven